

<!-- STREAMING_CHUNK:Structuring the project directory tree -->

DC Materials Vault: Modular Directory Tree & Architecture Specification

Below is the complete architectural directory tree separating the monolithic implementation into modular ES modules (type="module"), componentized stylesheets, and semantic HTML templates. Every data flow, hardware wedge listener, dual-directory storage pipeline, and Chart.js analytics engine is mapped 1-to-1 to guarantee zero feature regression.

```
dc-materials-vault/
├── index.html                    # Root HTML Shell & Layout
Mount Points
|
├── css/                          # Modular Style Sheets
|   ├── base.css                 # Font configurations, root
|   │   variables, scrollbar styles
|   ├── components.css          # Buttons, badges, cards,
|   │   input styles, toast animations
|   └── print.css                # Dedicated print stylesheet
|   for Code128 bin & SKU tags
└── js/                          # Pure Native ES6+ Modules
    ├── app.js                  # Master controller, view
    router, and lifecycle orchestrator
    ├── config/
    │   └── defaults.js          # Default catalog items,
    │   initial locations, & sample events
    ├── state/
    │   └── state.js             # Central reactive state
    object & subscriber notification bus
    ├── services/
    │   ├── storageService.js    # File System Access API &
    │   │   dual-directory persistence engine
    │   ├── scannerService.js    # Zebra HID hardware scanner
    │   │   wedge listener (<60ms cadence)
    │   └── barcodeService.js    # JsBarcode wrapper & Code128
    │   SVG label generator
    ├── models/
    │   ├── catalogModel.js      # SKU rules, packaging tiers
    │   │   (Item, Carton, Box), vendor UPCs
    │   └── locationModel.js     # Site hierarchy (Site > Room
    > Zone > Bin)
```

```

    |   └─ inventoryModel.js           # Stock balance matrix, event
reconciler, & consumption metrics
    |
    └─ components/
        └─ toast.js                   # Floating feedback
notifications (success, error, info)
    |   └─ printModal.js              # Label preview, print dialog
lifecycle, & SVG rendering
    |   └─ chartManager.js            # Chart.js velocity trendline
(Net Qty, Received, Consumed)
    |
    └─ views/
        └─ dashboardView.js           # Metrics, inventory balance
table, & category filters
        └─ catalogView.js             # SKU registry form,
packaging directory, & label triggers
        └─ locationView.js            # Storage bin hierarchy
builder & location barcode actions
        └─ massMoveView.js            # Staging manifest, bulk
transfer logic, & location validation
        └─ ledgerView.js              # Append-only transaction
log, CSV export, & Shift Digest

```

<!-- STREAMING_CHUNK:Specifying HTML and Stylesheet components -->

Detailed File & Module Specifications

1. Root & CSS Layer

index.html

- **Role:** The application host document. Loads external CDNs (Tailwind CSS, JsBarcode, Chart.js, and Google Fonts) and links stylesheets.
- **Containers:**
 - <header>: Brand badge, active scanner status pulse, view navigation buttons, and technician ID synchronization.
 - #scannerBanner: Fast-action popover triggered by hardware barcode scans with one-tap actions (Consume, Store, Receive across Unit, Carton, and Box tiers).
 - #subHeader: Work zone selector dropdown and manual simulation barcode input.
 - <main id="viewContainer">: Target outlet where modular views dynamically mount.
 - #printModalRoot: Isolated modal root designed to swap visibility during browser print triggers.
 - #toastNotification: Fixed-position notification card.
- **Script Loading:** Mounts <script type="module" src="js/app.js"></script>.

css/base.css

- **Role:** Global resets, typography defaults (Inter & JetBrains Mono), color tokens, and custom scrollbar definitions.
- **Responsibilities:**
 - Defines dark-mode palette variables (bg-slate-950, bg-slate-900, text-slate-100).
 - Custom webkit-scrollbar styling (5px track, slate-800 thumb) to maintain a modern terminal feel.

css/components.css

- **Role:** Scoped animations and component visual enhancements.
- **Responsibilities:**
 - Toast transition classes (translate-y-20, opacity-0, pointer-events-none).
 - Pulse animation for the Zebra HID scanner status indicator.
 - Dynamic button gradients and custom border highlights for active table rows.

css/print.css

- **Role:** Hard-isolated media query stylesheet (@media print).
- **Responsibilities:**
 - Enforces visibility: hidden !important across all body elements except #printModalRoot and .printable-barcode-card.
 - Configures page-break rules (page-break-inside: avoid), standard label dimensions (90mm), and high-contrast black/white rendering for thermal or laser label printers.

<!-- STREAMING_CHUNK:Defining State and Services modules -->

2. State & Core Services Layer (js/config/, js/state/, js/services/)

js/config/defaults.js

- **Role:** Supplies initial seed data if the local or shared storage folders are empty on first startup.
- **Exports:**
 - DEFAULT_CATALOG: Seed items (cabling, SFP+ optics, flag labels, combo locks, blanking panels) with predefined packaging rules (unitsPerCarton, cartonsPerBox, minThreshold).
 - DEFAULT_LOCATIONS: Hierarchical bins (LOC-INBOUND-DOCK, LOC-HALL1-A01, LOC-STOR-OPT01, etc.).
 - SEED_EVENTS: Initial historical receiving and consumption transactions to populate trend charts on first run.

js/state/state.js

- **Role:** Single source of truth for runtime variables.
- **Exports & Properties:**
 - state.activeView: Currently rendered tab (dashboard, catalog, locations, massMove, ledger).
 - state.techId: Currently logged technician badge (e.g., TECH-774).
 - state.activeLocationId: Active floor work zone (e.g., LOC-HALL1-A01).

- state.selectedCategory: Selected filter category (All, Fiber Cabling, etc.).
- state.manifestQueue: Array of staged materials for bulk relocation.
- state.storage: Directory handles for catalogs_local, materials_local, catalogs_shared, and materials_shared.
- state.catalog, state.locations, state.events: Arrays populated from storage or defaults.

js/services/storageService.js

- **Role:** Native File System Access API client and fallback synchronization manager.
- **Key Functions:**
 - connectDualStorage(): Triggers window.showDirectoryPicker() to acquire root access; creates or binds the 4 required subdirectories (catalogs_local, materials_local, catalogs_shared, materials_shared).
 - writeDualFile(localDirKey, sharedDirKey, filename, data): Concurrently writes payload strings to both local and shared directory handles.
 - appendAtomicMaterialEvent(eventData): Writes isolated transaction files named <timestamp>_<rand>_<TYPE>_<SKU>.json into materials_local/ and materials_shared/.
 - saveToLocalStorage() / loadFromLocalStorage(): Provides persistent browser caching whenever the native file system is disconnected.

js/services/scannerService.js

- **Role:** Intercepts high-speed hardware barcode scanner input.
- **Key Functions:**
 - initScannerListener(callback): Registers global keydown event listener.
 - Calculates character arrival deltas (timingThreshold < 60ms). Discards normal human typing while buffering high-velocity scanner streams.
 - On Enter, trims the buffered string and emits it to callback(scannedString).

js/services/barcodeService.js

- **Role:** Encapsulates the JsBarcode library for on-demand Code128 SVG rendering.
- **Key Functions:**
 - generateBarcodeSvg(svgElement, barcodeValue, options): Formats and draws vector barcodes with standard check digits, human-readable text labels, and configurable heights/margins.

<!-- STREAMING_CHUNK:Defining Business Logic and Domain Models -->

3. Business Models Layer (js/models/)

js/models/catalogModel.js

- **Role:** Enforces business logic for materials, packaging ratios, and barcode resolution.
- **Key Functions:**
 - createCatalogItem(itemData): Validates unique SKU constraints, sets unit of measure (UOM), carton/box multipliers, and writes updates to catalogs_local/ and

- catalogs_shared/.
- findItemByBarcode(barcodeStr): Searches across internal SKUs and vendor OEM barcodes.

js/models/locationModel.js

- **Role:** Manages facility, room, and bin hierarchy.
- **Key Functions:**
 - createLocation(locationData): Registers new site positions (site, room, zone, barcode), verifies uniqueness, and saves to catalogs_local/ and catalogs_shared/.
 - findLocationByBarcode(barcodeStr): Identifies if a scanned tag corresponds to an active storage bin.

js/models/inventoryModel.js

- **Role:** Reconciles the event stream into balance sheets and velocity aggregates.
- **Key Functions:**
 - calculateStockMatrix(): Replays all events (RECEIVED, STORED, CONSUMED, MASS_RELOCATE) to produce net quantities by location, along with isolated consumption and receiving historical timelines.
 - getMetricInWindow(historyArray, hours, locationFilter): Computes rolling window metrics (e.g., 24h Received, 24h Consumed, 7d Consumed) filtered by location.
 - recordQuickAdjustment(sku, delta, actionType, packagingTier, reason): Converts carton/box quantities into unit base counts and emits atomic events through storageService.

<!-- STREAMING_CHUNK:Defining UI Components and View Controllers -->

4. Components & Views Layer (js/components/, js/views/)

js/components/toast.js

- **Role:** Displays floating feedback alerts without blocking the floor technician's workflow.
- **Exports:**
 - showToast(message, type = 'info' | 'success' | 'error'): Modulates color themes and handles automated timeout dismissal.

js/components/printModal.js

- **Role:** Modal dialog lifecycle for previewing and printing Code128 labels.
- **Exports:**
 - openPrintModal(code, title, tagType): Mounts label details, calls barcodeService, and exposes standard print controls.
 - closePrintModal(): Dismisses modal overlay.

js/components/chartManager.js

- **Role:** Manages the Chart.js lifecycle and dynamic dataset updates.
- **Exports:**

- `initInventoryChart(canvasElement)`: Instantiates the line graph with custom dark-slate axes, tooltip styling, and multi-line dataset bindings.
- `updateInventoryChart(itemFilter, locFilter, timeWindow)`: Replays historical event timelines into stepped time labels, plotting **Net Quantity Balance**, **Cumulative Received**, and **Cumulative Consumed**.

js/views/dashboardView.js

- **Role:** Renders View 1 (Macro metrics, interactive trendline canvas, category filter pills, and live on-hand balance table).
- **Key Functions:**
 - `renderDashboard(container)`: Assembles markup, attaches rapid adjustment buttons (-1, +1, +Ctn), and initializes chart filters.

js/views/catalogView.js

- **Role:** Renders View 2 (SKU catalog registry form and packaging directory table).
- **Key Functions:**
 - `renderCatalog(container)`: Mounts the item registration form and catalog table with inline "Print Barcode" action buttons.

js/views/locationView.js

- **Role:** Renders View 3 (Location registration form and registered bins directory).
- **Key Functions:**
 - `renderLocations(container)`: Mounts hierarchy inputs and renders location barcode printing cards.

js/views/massMoveView.js

- **Role:** Renders View 4 (Bulk relocation manifest builder).
- **Key Functions:**
 - `renderMassMove(container)`: Handles source-to-target dropdowns, live stock availability validation, staging list additions, and batch commitment dispatch.

js/views/ledgerView.js

- **Role:** Renders View 5 (Append-only audit ledger).
- **Key Functions:**
 - `renderLedger(container)`: Renders event history table, triggers CSV snapshot downloads, and generates the copy-to-clipboard 12-hour Shift Material Logistics Digest.

<!-- STREAMING_CHUNK:Defining Application Controller and execution lifecycle -->

5. Application Orchestrator (js/app.js)

js/app.js

- **Role:** Main bootstrap and dependency wiring script.
- **Execution Flow:**
 1. Imports models, services, views, and components.
 2. Runs loadFromLocalStorage() or prompts for connectDualStorage().
 3. Registers DOM event listeners for navigation tabs (switchView), technician badge updates, and active location selectors.
 4. Starts scannerService with the central barcode resolution handler:
 - If barcode matches a location \to Locks work zone.
 - If barcode matches an SKU/UPC \to Opens quick-action bar with calculated Unit/Carton/Box buttons.
 - If barcode is unrecognized \to Prompts new SKU cataloging with auto-filled code.
 5. Initializes the default view (dashboard) and boots chartManager.